

APPLIED ARTIFICIAL INTELLIGENCE

Minor in Applied Artificial Intelligence

Applications of Artificial Intelligence (AI) are permeating nearly every aspect of society, from healthcare diagnostics, financial analysis, and transportation optimization to enhancing educational methodologies, driving innovations in entertainment, and shaping more efficient and personalized customer experiences. The Applied AI minor is available to all undergraduate students at Iowa State University. It provides foundational knowledge on the use of AI using low-code or no-code tools available to a wide range of users. A useful analogy is that of being able to drive a car versus produce it or perform work under the hood. The minor also helps students understand the ethical, social, cultural, and economic implications of increased AI use.

The minor in Applied Artificial Intelligence requires the completion of 15 credit hours, including a 6 credit core plus 9 additional credits selected from the approved list.

Students with strong backgrounds in logical or quantitative reasoning who are interested in learning basic AI and machine learning techniques and applying AI methods to solve practical problems may be interested in the Artificial Intelligence Undergraduate Minor (<https://catalog.iastate.edu/collegeofliberalartsandsciences/artificialintelligence/>).

MINOR

The minor in Applied Artificial Intelligence requires the completion of 15 credit hours, including a 6 credit core plus 9 additional credits selected from the approved list.

AI 2010	Introduction to Applied AI	3
AI 2020	Ethical Design, Use, and Impact of AI	3
Choose 9 additional credits from the list below		9
ABE 5160	Data Science and Analytics for Agricultural and Biosystem Engineers	
ANTHR 3160X	Anthropology and Artificial Intelligence (AI)	
ART 3400X	Artificial Intelligence (AI) in Art and Design	
COMS 4720	Principles of Artificial Intelligence	
COMS 4740	Introduction to Machine Learning	
COMS 4760	Motion Planning for Robotics and Autonomous Systems	
COMST 3300	Communication in a Digital World	
DS 2010	Introduction to Data Science	
DS 3010	Applied Data Modeling and Predictive Analysis	

EDUC 4540 Specified Section	Emerging Topics in Educational Technology - Only topic of Introduction to AI in Education
EE 4250	Machine learning: A Signal Processing Perspective
EE 4280	Image Analysis using Machine Learning
ENGL 1200	Language and Technology
ENGL 2220	Artificial Intelligence and Writing
ENGL 3190 Specified Section	Studies in Language Use - Only topic of Human Language and Generative AI
ENGL 4100	Language as Data
IE 4830	Data Mining
IE 4870	Big Data Analytics and Optimization
JLMC 4740	Communication Technology and Social Change
MATH 4220X	Mathematical Principles of Data Science
ME 2800	Design and Analysis of Cyber-Physical Systems
MIS 4360	Introduction to Business Analytics
MIS 4460	Advanced Business Analytics
MKT 4560X	Digital Marketing Analytics
PHIL 3430	Ethics of Computing and Artificial Intelligence
PHIL 4650	Brains, Minds, and Computers
PSYCH 3500	Human Factors in Technology

At least 6 credits must be in courses numbered 3000 or above. At least 3 credits must be taken at Iowa State University.

At least 3 credits used for the minor cannot be used to meet any other department, college or university requirement.

Courses taken for a minor may not be taken on a pass-not pass basis.

The Applied Artificial Intelligence Undergraduate Minor is an LAS Minor. In addition to University policies governing minors (<https://catalog.iastate.edu/academics/#degreeplanningtext>), LAS minors require at least 6 credits in courses numbered 3000 and above, with a grade of C or higher.

Other courses not currently listed may be approved to count towards additional credits. Contact the Applied Artificial Intelligence minor professor-in-charge for information on eligible courses.